

Erdős Problem #796

EP-796

L1 - TRACTABLE

PARTIALLY SOLVED

Number Theory

ORIGINAL PROBLEM STATEMENT

Let $k \geq 2$ and let $g_k(n)$ be the largest possible size of $A \subseteq \{1, \dots, n\}$ such that every m has $< k$ solutions to $m = a_1 a_2$ with $a_1 < a_2 \in A$.

Is it true that $g_3(n) = \frac{\log \log n}{\log n} n + (c + o(1)) \frac{n}{\log n}$ for some constant c ?

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Archive: <https://www.unsignedmath.com/problems?difficulty=1>

Dataset: <https://huggingface.co/datasets/ulamai/UnsignedMath>